Rational Bases and the Integer-Base Argument

A Lean-checked strategy separation for Erdős Problem #1049

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Abstract

Erdős's argument for the irrationality of $\sum_{n\geq 1} c(n)b^{-n}$ at an integer base $b \geq 2$ proceeds by clearing denominators at a cut level and trapping a positive integer below 1. At a rational base r/s with $s \geq 2$ the same clearing leaves a factor s^{N+1} behind. We formalise in Lean the elementary consequence of that residue. Writing a coordinatewise corridor for the surviving arithmetic, we prove that a corridor forces $s^{N+K+1} < r(N+K)$, and that at the base $3/2$ this is impossible once the shift and the cleared window are both nonempty. We also prove the exact denominator-cleared tail recurrence $U_{N+1} = rU_N - Bc(N+1)s^{N+1}$ and the bound $2^{N+1} \leq Bc(N+1)s^{N+1}$ for $s \geq 2$ and positive data, against the exact collapse of that term to $Bc(N+1)$ at $s = 1$.

Problem #1049 is open, and nothing here decides it. The results are statements about one clearing scheme, not about the arithmetic nature of any value: we do not prove irrationality at $3/2$, at any other rational base, or in the unrestricted problem. The published height criterion that does settle a restricted family of rational bases, including $7/2$, is used as an external theorem; of its inputs we formalise only two exact arithmetic facts, and we state plainly which three inputs are not formalised. Separately we check polynomial bounds on the denominator exponents of a homogenised Padé construction, with the analytic remainder left untouched. For the stronger Heine–Zudilin route we isolate a sharper obstruction. Rowwise content scales the analytic error and the exterior determinant by exactly the same factors, including the absolute determinant height. Moreover homogeneous specialisation at $(3, 2)$ sees the constant endpoint modulo 3 and the top endpoint modulo 2, so unit endpoints exclude both primes from any common multiplier. Thus the missing local gain must survive primitive normalisation. The surviving programme is a growing-rank additive kernel that cancels both endpoint jets in both primitive coefficient channels while preserving a nonzero, sufficiently small remainder. We count the four-jet target exactly and prove the concrete sufficient threshold $n \geq 4R + 2S$ at positive bottom depth.

1 The problem

Let $t > 1$ be a rational number and let $\tau(n)$ count the divisors of n. Erdős Problem #1049 asks whether

$$F(t) = \sum_{n\geq 1} \frac{1}{t^n - 1} = \sum_{n\geq 1} \frac{\tau(n)}{t^n}$$

Authorship and AI use. The prose in this version was generated by large-language-model agents under Will Cook's direction. Cook set the objectives and acceptance criteria, reviewed and authorised the manuscript, and is responsible for its contents. The agents are production tools, not authors. See *Statements and declarations*.

is irrational [2, p. 102]. The two forms agree by expanding $(t^n - 1)^{-1} = \sum_{k\geq 1} t^{-nk}$ and collecting the terms with the same exponent, the coefficient of t^{-m} being the number of divisors of m. The question is a conjecture of Chowla; Erdős proved it for every integer $t \geq 2$ [1]. Numbering and status follow Bloom's catalogue [7]. For non-integer rational t the problem is open.

Write $t = r/s$ in lowest terms with $r > s \geq 1$, so that $s = 1$ is exactly the integer case Erdős settled. The smallest resistant explicit base is $t = 3/2$. A published height criterion of Bundschuh and Väänänen [3] settles a family of rational bases restricted by a height condition; that family contains $7/2$ and does not contain $3/2$. Between the two lies the question this note is about: what exactly stops the integer-base argument from running at $3/2$?

Relation to prior work. The statement of Problem #1049 has been formalised before, as a conjecture with an unfilled proof, in the *Formal Conjectures* collection [8]. That is a statement of the question; the present note formalises propositions about the clearing argument, and neither answers it. Erdős's positive-integer theorem [1] sits inside a larger integer-base literature: Van Assche recovered the integer-base irrationality and an irrationality-measure bound using little q-Legendre Padé approximants [5, Thm. 1], while Vandehey completed the elementary digit method for negative integer bases and bounded coefficient sequences [6, Thm. 1.2]. Both retain an integer base. The rational non-integer progress relevant here is instead the height criterion of Bundschuh and Väänänen [3]. This makes the primitive-normalisation no-go and the surviving additive-kernel programme, rather than the elementary corridor observation alone, the main contribution to surface from this note. No claim of priority is made for anything below, which concerns the formal status of an argument rather than a new theorem about $F(t)$.

The integer-base argument, in outline. Suppose $F(b) = p/q$. Cut the series at a level N, multiply by qb^N, and subtract the cleared prefix. What remains is a positive integer, since every term of the prefix has been cleared and the tail is positive; and it is smaller than 1, since the tail of a geometric-type series is small. A positive integer below 1 does not exist. The argument is elementary and it is sensitive in exactly one place: the clearing must remove every denominator.

At $\beta = r/s$ the term $c(n)\beta^{-n}$ is $c(n)s^n/r^n$. Clearing the power of r leaves the numerator factor s^n in place. That factor is invisible when $s = 1$ and grows geometrically when $s \geq 2$. The two sections that follow record what that residue costs, first as a no-go for the clearing scheme (Section 2) and then as an exact recurrence with a lower bound on the surviving term (Section 3).

Status of everything below. *Status.* The problem treated here is open, and this note does not close it. Every statement below marked as checked is a proposition that the pinned Lean kernel accepts from the sources this note links to, with no `sorry`, no added axiom, and no unchecked evaluation. That is a claim about the formal statement, not about its mathematical interest, its novelty, or the original problem. The unresolved obligations are named exactly, in their own section, and none of the finite computations, reductions, or no-go results here removes one of them.

Statement	Status	Treatment here
Irrationality of $F(3/2)$	Open	Not proved, and no partial result here bears on it.
Irrationality at some rational non-integer base	Proved elsewhere	The height criterion of [3]; cited, not formalised.
Corridor forces $s^{N+K+1} < r(N+K)$	Proved here	Theorem 2.2.
No corridor at base $3/2$	Proved here	Theorem 2.4; a statement about the clearing scheme.
Cleared-tail recurrence	Proved here	Theorem 3.1, an exact identity.
Forcing term at least 2^{N+1} for $s \geq 2$	Proved here	Theorem 3.2; at $s = 1$ the term is $Bc(N+1)$.
Two arithmetic facts behind the $7/2$ criterion	Proved here	Section 4; three further inputs are not formalised.
Padé denominator-exponent bounds	Proved here	Section 5; exponent arithmetic only.
Padé remainder positivity and decay	Not treated	Section 5.
Homogeneous endpoint residues at $(3,2)$	Proved here	Theorem 7.1.
Scalar-cone product-formula margin	Negative under source input	Theorem 7.3; the source inequality is assumed.
Growing-rank simultaneous endpoint-jet kernel	Open	Section 8; the exact surviving producer.

Reading map. Sections 2 and 3 are the content; Sections 4 and 5 record what is and is not formalised of two external routes. Section 8 states the remaining obligations. Linked phrases open the corresponding Lean declaration at the pinned source revision d7c339fe1e2c.

Keywords. irrationality; Lambert series; rational base; Padé approximation; Lean 4.
MSC 2020. 11J72 (primary); 11J82, 68V20 (secondary).

2 The coordinatewise corridor

We isolate the arithmetic that the clearing scheme leaves behind, in a form that does not mention the series.

Definition 2.1 (coordinatewise corridor)**.** Let a, b, N, K, Q, D be natural numbers. Say that (a, b, N, K, Q, D) is a *coordinatewise corridor* when

$$a > 0, \qquad Q > 0, \qquad D > 0, \qquad D \leq N + K, \qquad a^K \mid QD, \qquad Q\, b^{N+K+1} < a^{K+1}.$$

The reading is: a and b are the numerator and denominator of the base, N is the shift, K is the width of the cleared window, Q is the accumulated clearing factor, and D is the final coefficient being cleared. The bound $D \leq N + K$ is the only property of the coefficient used; for the divisor-counting coefficient it holds because $d(n) \leq n$. The divisibility $a^K \mid QD$ is the requirement that clearing succeeded coordinatewise, and the last inequality is the tail estimate that makes the trapped integer smaller than 1. The definition is the corridor predicate.

Theorem 2.2 (corridor bound). *If (a, b, N, K, Q, D) is a coordinatewise corridor, then*

$$b^{N+K+1} < a\,(N + K).$$

Proof. Since $QD > 0$ and $a^K \mid QD$, we have $a^K \leq QD$, and $D \leq N+K$ gives $a^K \leq Q(N+K)$. Hence

$$Q\,b^{N+K+1} < a^{K+1} = a^K \cdot a \leq Q(N + K) \cdot a,$$

and cancelling the positive factor Q gives the claim. □

Formalised as the power-versus-linear consequence.

The inequality of Theorem 2.2 is where the integer and rational cases part. At $b = 1$ it reads $1 < a(N + K)$, which holds for every $a \geq 2$ and every nonempty window; the corridor imposes no obstruction at all. At $b \geq 2$ the left side is exponential in $N + K$ and the right side is linear, so the corridor can survive only for small $N + K$. At $b = 2$ the crossing has already happened at the smallest admissible window.

Proposition 2.3. *For every natural number $x \geq 2$ we have $3x < 2^{x+1}$.*

Proof. Induction from $x = 2$, where $6 < 8$. For the step, $2^{x+1} \geq 2^2 > 3$ when $x \geq 1$, so $3(x + 1) = 3x + 3 < 2^{x+1} + 2^{x+1} = 2^{x+2}$. □

Formalised as the exponential comparison.

Theorem 2.4 (no corridor at base 3/2). *For all $N \geq 1$ and $K \geq 1$ and all natural Q, D, the tuple $(3, 2, N, K, Q, D)$ is not a coordinatewise corridor.*

Proof. A corridor would give $2^{N+K+1} < 3(N + K)$ by Theorem 2.2, contradicting Proposition 2.3 applied to $x = N + K \geq 2$. □

Formalised as the corridor exclusion at three halves.

What this section proves, and what it does not. Theorem 2.4 is a statement about Definition 2.1 and about nothing else. It says that the arithmetic left over by a literal coordinatewise clearing at base 3/2 is inconsistent, and therefore that the integer-base argument does not transfer in that form. It does not bound the denominator of $F(3/2)$, it does not show that $F(3/2)$ is irrational, and it does not show that $F(3/2)$ is rational. It also does not rule out a clearing scheme of a different shape: the corridor fixes one divisibility pattern and one tail inequality, and an argument organised differently is not covered by the hypothesis.

3 The cleared tail and the denominator tax

Theorem 2.4 says that one scheme fails. This section identifies the quantity responsible, as an exact recurrence.

Throughout, r, s, B, F are rationals with $r \neq 0$, and $c : \mathbb{N} \to \mathbb{Q}$ is arbitrary. Define the prefix and the cleared tail state by

$$P_N = \sum_{m=0}^{N-1} \frac{c(m+1)\, s^{m+1}}{r^{m+1}}, \qquad U_N = B\, r^N (F - P_N).$$

Thus P_N is the partial sum of $\sum_{n\geq 1} c(n)(s/r)^n$ through level N, and U_N is the tail of a putative value F after that level, scaled by Br^N. These are the rational-base prefix and the cleared tail state.

Theorem 3.1 (cleared-tail recurrence)**.** *For every N,*

$$U_{N+1} = r\, U_N - B\, c(N+1)\, s^{N+1}.$$

Proof. Expanding $P_{N+1} = P_N + c(N+1)s^{N+1}/r^{N+1}$ and $r^{N+1} = r^N \cdot r$ in the definition of U_{N+1} and clearing the denominator r^{N+1}, which is nonzero, gives the identity. □

Formalised as the cleared-tail recurrence.

The recurrence has a linear part rU_N and a forcing term $Bc(N+1)s^{N+1}$. The integer-base argument works by keeping the state in a bounded window, and the forcing term is what it must absorb at each step. Its size is exactly the point of departure.

Theorem 3.2 (the forcing term)**.** *Let s, B be natural numbers and $c : \mathbb{N} \to \mathbb{N}$, and put $G_N = B\, c(N+1)\, s^{N+1}$.*

1. *If $s \geq 2$, $B \geq 1$ and $c(N+1) \geq 1$, then $2^{N+1} \leq G_N$.*
2. *If $s = 1$, then $G_N = B\, c(N+1)$.*

Proof. For the first part, $2^{N+1} \leq s^{N+1} = 1 \cdot s^{N+1} \leq Bc(N+1)s^{N+1}$, using $B\, c(N+1) \geq 1$. The second part is the definition with $s = 1$. □

Formalised as the exponential lower bound and the integer-base collapse.

Reading the two parts together. At $s = 1$ the forcing term is $Bc(N+1)$, so it grows only as fast as the coefficient; for the divisor function that is $O(N^{\varepsilon})$ for every $\varepsilon > 0$, and a bounded-state argument has room. At $s \geq 2$ the same term is at least 2^{N+1} whenever the coefficient is nonzero. This is an exact lower bound on one quantity, and it is all that is proved. It is not a proof that no bounded-state argument exists at $s \geq 2$; the theorem that one particular scheme fails is Theorem 2.4, and the bound here records the size of the term that scheme would have to absorb.

4 The height criterion at $7/2$

Bundschuh and Väänänen [3] prove an irrationality criterion for a family of rational bases cut out by a height condition. The criterion applies at $\beta = 7/2$. It is an analytic theorem and it is not formalised here. Two of its elementary inputs are.

Proposition 4.1. *$2^{18} < 7^7$, and $\frac{7}{18} = \frac{1}{2} - \frac{1}{9}$ in $\mathbb{Q}$.*

These are the integer power comparison, namely $262\,144 < 823\,543$, and the exact rational boundary.

The first is the arithmetic content of the estimate $\log 2/\log 7 < 7/18$; the passage from the integer inequality to the logarithmic one uses monotonicity of the logarithm, which is not part of the formalised statement. The second is the exact form of a margin whose other input is the strict bound $1/\pi^2 < 1/9$, that is $3 < \pi$, which is also not formalised here. Three inputs are therefore external: the logarithmic step, the bound on π, and the analytic theorem of [3] itself. We record the two exact arithmetic facts separately from the theorem precisely so that the two cannot be confused; a reader should not take Proposition 4.1 as a formalisation of the published criterion, because it is not one.

The criterion does not cover $3/2$, and the results of Sections 2 and 3 give no evidence either way about whether $F(3/2)$ is irrational.

5 Padé denominator exponents

A second external route replaces the clearing argument by explicit rational approximation. One constructs integer linear forms in 1 and $F(\beta)$ whose analytic decay outruns the height of their common denominator. For a homogenised construction over integer parameters the proposed common denominator exponent is $E_n = (3n^2 - n)/2$. Since only doubled exponents occur below, every statement lives over $\mathbb{Z}$ and no parity bookkeeping is needed; we write $\widetilde{E}_n = 2E_n = 3n^2 - n$.

Proposition 5.1 (summand bound and exact gap)**.** *Put*

$$\widetilde{P}(n,k) = 2(k(n-k) + nk) + k(k-1),$$

$$\widetilde{Q}(n,m) = 2(n^2 - n) + j^2 + 2jm + j - m^2 + 3m, \qquad j = n - m - 1.$$

Then, for integers n, k, m:

1. *if $0 \le k \le n$, then $\widetilde{P}(n,k) \le \widetilde{E}_n$, and the gap factors as $\widetilde{E}_n - \widetilde{P}(n,k) = (n-k)(3n-k-1)$;*
2. *$\widetilde{E}_n - \widetilde{Q}(n,m) = 2(n + m(m-1))$ identically;*
3. *if $n \ge 0$ and $m \ge 1$, then $\widetilde{Q}(n,m) \le \widetilde{E}_n$.*

Part (1) is the summand exponent bound, part (2) the exact gap identity, and part (3) the maximal exponent bound. The gap in (2) is an identity in $\mathbb{Z}[n,m]$, so (3) follows from $m(m-1) \ge 0$ and $n \ge 0$; the hypothesis $m \ge 1$ names the range in which the corresponding summand does not vanish, and it is not the sharpest hypothesis under which the inequality holds.

These are inequalities between polynomials in the exponents. They establish that the proposed common denominator is large enough to clear the construction, and nothing further. Positivity of the remainder, its rate of decay, and the comparison of that rate against the denominator height are the analytic obligations, and none of them is treated here. A reconstructed identity is not an irrationality measure until those are closed.

6 Primitive normalisation is compulsory

For an integer coefficient pair (U, V) and a real target S, put

$$L_S(U,V) = US - V, \qquad \Delta((U_n,V_n),(U_m,V_m)) = U_nV_m - U_mV_n.$$

The second expression eliminates S exactly:

$$\Delta = U_mL_S(U_n,V_n) - U_nL_S(U_m,V_m).$$

This is the determinant consumer for any two rational approximants.

Theorem 6.1 (row-content no-go)**.** *For integers c_n, c_m,*

$$L_S(c_nU_n, c_nV_n) = c_nL_S(U_n,V_n),$$

$$\Delta(c_n(U_n,V_n), c_m(U_m,V_m)) = c_nc_m\Delta((U_n,V_n),(U_m,V_m)),$$

and consequently

$$|\Delta(c_n(U_n,V_n), c_m(U_m,V_m))| = |c_n|\,|c_m|\,|\Delta((U_n,V_n),(U_m,V_m))|\,.$$

In particular c_nc_m divides the scaled determinant. Hence a local divisor supplied only by the two row contents is paid for by exactly the same factor in the Archimedean determinant height.

Proof. All three identities are direct expansions in $\mathbb{Z}$ or $\mathbb{R}$; the divisibility statement uses the primitive determinant as its witness. □

Lean checks the error identity in error scaling, the determinant identity in content factorisation, the exact absolute-height identity in absolute determinant scaling, and the divisor statement in content-product divisibility. The elimination identity is the checked exterior determinant identity.

The theorem does not construct primitive Padé rows, estimate their remainders, or prove that their exterior determinant is nonzero. It removes one tempting source of false gain: multiplying a useful row by a large common integer cannot improve the local-to-Archimedean balance. Every candidate family must first be divided by its rowwise common contents; the required 2-adic and 3-adic gain must remain afterwards.

7 The endpoint obstruction at $3/2$

The Heine–Zudilin forms have integral coefficient polynomials before specialisation. For $P(X) = \sum_i p_iX^i \in \mathbb{Z}[X]$ and a declared width W, put

$$H_W(P) = \sum_{i=0}^{W} p_i\,3^i2^{W-i}.$$

This is the homogeneous endpoint evaluation.

Theorem 7.1 (endpoint residues)**.** *For every integral polynomial P,*

$$H_W(P) \equiv p_0\, 2^W \pmod 3, \qquad H_W(P) \equiv p_W\, 3^W \pmod 2.$$

Consequently a unit constant coefficient prevents divisibility by 3, and a unit top coefficient prevents divisibility by 2.

Proof. Modulo 3, every summand with $i > 0$ vanishes; modulo 2, every summand with $i < W$ vanishes. The remaining powers are units in the corresponding residue fields. □

The congruences are the bottom-endpoint identity and top-endpoint identity; the unit consequences are the constant-endpoint obstruction and top-endpoint obstruction.

There is a useful common-multiplier form of the same boundary. If U has unit top endpoint, V has unit constant endpoint, and an integer c divides both $H_W(U)$ and $H_W(V)$, then

$$2 \nmid c \qquad \text{and} \qquad 3 \nmid c.$$

This is the checked common-multiplier exclusion. Together with Theorem 6.1, it says that neither rowwise scalar content nor a common multiplier of the specialised channels can manufacture the two local primes required at $3/2$ under the stated endpoint hypotheses.

For higher powers define the bottom and top jets by reducing $H_W(P)$ modulo 3^R and 2^S, respectively. Their vanishing is exactly the requested divisibility:

$$J_{3,R}(P) = 0 \iff 3^R \mid H_W(P), \qquad J_{2,S}(P) = 0 \iff 2^S \mid H_W(P).$$

These are the checked bottom-jet consumer and top-jet consumer. They turn the missing local divisor into an additive congruence kernel problem.

Theorem 7.2 (binary four-jet collision)**.** *Let $(U_j, V_j)_{j<n}$ be any n integral coefficient pairs. If the 2^n binary selectors outnumber the finite four-jet target*

$$(\mathbb{Z}/3^R\mathbb{Z})^2 \times (\mathbb{Z}/2^S\mathbb{Z})^2,$$

then two distinct subsets have the same four-jet sum. Subtracting their indicator vectors gives a nonzero coefficient vector in $\{-1, 0, 1\}^n$ cancelling all four jets. The target has exact cardinality

$$(3^R)^2(2^S)^2.$$

In particular, if $R > 0$ and $4R + 2S \le n$, such a collision exists.

Proof. Send each binary selector to the sum of the four-jet signatures it selects. The claimed cardinal inequality and the pigeonhole principle give two distinct selectors in the same fibre. The cardinality formula is the product of the four cyclic-modulus cardinalities. For $R > 0$,

$$(3^R)^2(2^S)^2 < (4^R)^2(2^S)^2 = 2^{4R+2S} \le 2^n,$$

which proves the stated sufficient threshold. □

The target count is the checked four-jet target cardinality; the abstract collision is the checked four-jet pigeonhole kernel, and the linear sufficient condition is the checked rank–depth collision threshold. Pigeonhole cancellation itself requires no independence. Independence or a non-collapsed deformation is needed only to ensure that the resulting nonzero selector difference has a nonzero polynomial pair and analytic remainder.

Theorem 7.3 (scalar margin no-go)**.** *Let $C_1 > 0$. If $C_0 \le 0$ or $2C_0 \le C_1$, then*

$$C_0 \log 3 - C_1 \log 2 < 0.$$

This is the checked three-halves scalar margin. The primary Zudilin theorem [4] supplies an integer-base irrationality-exponent estimate on its parameter cone, and the elementary inequality $\mu \ge 2$ then forces $2C_0 \le C_1$ whenever $C_0 > 0$; Lean checks that implication separately. The primary theorem assumes an integer base $p = 1/q$. It does not state a rational $p = 3/2$ theorem, so the all-scale coefficient construction and rational specialisation remain external to the checked result. What is closed here is the scalar parameter margin, not Problem #1049.

8 Complements and further questions

Problem #1049 is open. The current 3/2 frontier is the following exact producer.

Problem 8.1 (growing-rank simultaneous endpoint-jet kernel)**.** For positive quadratic target depths R_n, S_n, exhibit at least $4R_n + 2S_n$ *primitive-normalised* Heine–Zudilin coefficient pairs from a genuinely non-collapsed deformation, and prove that at least one resulting checked collision satisfies

$$J_{3,R_n}\Big(\sum_j \lambda_j U_j\Big) = J_{3,R_n}\Big(\sum_j \lambda_j V_j\Big) = 0, \qquad J_{2,S_n}\Big(\sum_j \lambda_j U_j\Big) = J_{2,S_n}\Big(\sum_j \lambda_j V_j\Big) = 0,$$

where R_n, S_n have the required quadratic order, while the resulting remainder is nonzero and its local gain beats the Archimedean height. The checked collision already has coefficients in $\{-1, 0, 1\}$. A rank-saturated contiguous-shift family or a genuinely independent deformation is an acceptable first producer.

Problem 8.2 (the published region)**.** Formalise the elementary parameter inequalities of the criterion of [3], keeping the analytic theorem external and cited. Proposition 4.1 formalises two of its inputs; the logarithmic monotonicity step and the bound $3 < \pi$ remain.

The unrestricted question, which rational bases give an irrational value, remains open, and is not reduced to either problem above.

Wanted. At positive target depths R_n, S_n, an explicit family of at least $4R_n + 2S_n$ primitive coefficient pairs, plus a proof that some checked collision has nonzero polynomial pair, nonzero remainder, and sufficient local gain after primitive normalisation. A scalar ray, row-content growth, a multiplicative cyclotomic factor, or another common-multiplier computation is not responsive to the surviving obstruction.

Already checked. The coordinatewise corridor fails at 3/2; the cleared-tail forcing term has an exponential denominator tax; the displayed Padé denominator exponents clear their finite sums; unit endpoints prevent common factors 3 and 2; row contents factor exactly from errors and determinants and pay the same factor in absolute determinant height; under unit endpoints a common specialised multiplier contains neither 2 nor 3; the source integer-base ratio inequality implies a strictly negative scalar margin at 3/2; and a large enough binary selector space forces a nonzero $\{-1, 0, 1\}$ four-jet collision. The exact target size is $(3^R)^2(2^S)^2$, and $n \geq 4R + 2S$ suffices when $R > 0$. None of these statements produces a sufficiently large primitive non-collapsed family or proves a collision polynomial pair or remainder nonzero.

Most useful answer. Give the first actual primitive-normalised family with $r(n) \geq 4R_n + 2S_n$, prove that the required local gain survives that normalisation, then show that its collision fibre is not contained in the polynomial-pair or remainder nullspace. If every available family is too small or degenerate, a rank theorem identifying that obstruction is more valuable than further low-rank enumeration.

If answered. A positive construction supplies the missing *primitive* local-gain producer to the existing exterior-determinant consumer. A negative rank theorem removes the additive endpoint route and exposes the next genuinely different deformation.

Statements and declarations

Artefact and data availability. The pinned formal-source revision contains the Lean sources, the fixed toolchain, and the library manifest used in the verification. This manuscript provides navigation rather than proof authority.

Declaration of generative AI use. Large-language-model agents were used throughout development to draft and revise prose, formal proofs, and software. The author set the objectives and acceptance criteria, selected and reviewed the claims, and approved the published version. The author assumes responsibility for the accuracy, interpretation, and presentation of the work. Generative systems are production tools; they are not authors and supply no independent authority. Lean checks each proof term against the fixed library version, and the sources linked here contain no proof placeholders and no project-defined axioms; Lean does not authorise the exposition, the citation choices, or the interpretation, for which the author remains responsible.

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A Guide to the formal sources

Each linked phrase opens its Lean declaration at the pinned source revision d7c339fe1e2c. The declarations of this note live in three modules: the corridor, the cleared-tail recurrence, and the two arithmetic certificates in the first; the Padé exponent arithmetic in the second; and the Heine–Zudilin endpoint arithmetic in the third. The link coordinates are validated against that pinned revision, so they remain correct as later work moves lines in the working tree.

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